FuluFhedzani

1. A database - Is a structured digital storage system where data is stored In a way that makes it easy to find, sort and use whenever needed.

Examples: ① Banks - They use databases to store or keep Customer information and transaction records

② Universities - They use databases for Students records, Grades & Courses AS well as financial System eg tuition fees, Scholarship

2. Data warehouse - Is a large storage System designed to collect & Store data from different Sources in one central place for Analysis & Reporting

* A database is used for daily operations & real-time transactions, while data Warehouse is used for analyzing large amounts of historical data.

* A company would use data warehouse to analyse customer's spending patterns and Generate financial Reports

3. Data lake - Is a system that stores large amounts of raw data in its original format; it can be structured (tables), semi structured or unstructured (images, videos, Emails)

* It does not organise data strictly before storing it, data is stored first and only processed or analyzed when needed; whereas datawarehouse Organised data before it is stored.

* A business would Prefer to use a data lake when it needs to store large amounts of raw, diverse data and is not yet sure how data will be analyzed, especially when the data is unstructured or mixed (videos, images, Social Media posts)

4. Data Mart - It is a smaller, specific version of a data warehouse that is department or business area specific within an Organization

- it is usually used to Support particular functions like Sales, finance or marketing

5. a, Structured - highly organized + follows a fixed format, usually stored in rows and columns, which is easy to search & Analyse eg Spreadsheet

b, Semi-Structured - Does not fit neatly into tables but still has some organization through tags or markers that help define elements eg JSON file with information like names, emails & purchase history

c, Unstructured - has no fixed format or Organization making it harder to store & Analyse directly eg A video recording, Emailtext, Social Media Image

6. Data modelling - Is the process of designing how data is organized, stored and related within a system or database. It shows how different data connect to each other

* It helps structure data in a clear and logical way, makes data easy to build and Maintain, Improves data consistency and reduces errors

7. Types 1. Online Shopping (Bash, takealot)

Bash
- past purchases
- Item in cart
- most purchased items

→ Recommend products a customer is likely to buy eg Sunscreen after buying facelotion

ABSA Bank
- Transaction Patterns
- Amount usually spend on online Purchases
- maximum deposit Patterns
→ It can detect freudulent activity by seeing unusual spending.

7. Data mining - Is the process of examining large sets of data do discover Patterns, trends and useful information that are not always obvious

8. Database Management Systems (DBMS) - Is a software that is used to Create, Store, Organise, Manage and retrieve data from a database, it is a middle layer between the user and the database, allow users to add, update, delete & retrieve data

a, MySQL - Is used for web application, stores data in tables & uses SQL

b, PostgreSQL - Advanced analytics and complex data

c, MongoDB - stores data in flexible, JSON like-documents instead of tables.
- Use real-time application & big data systems such as mobile apps

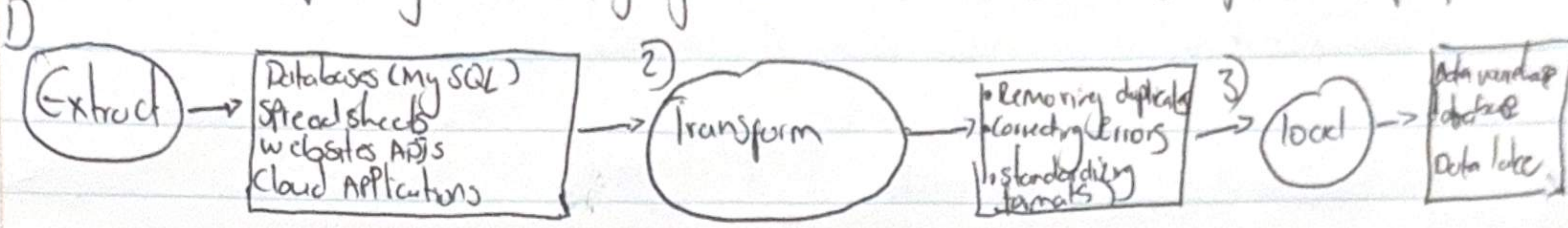
d, Oracle - Used by in realtime application by large Organizations like banks and insurance company for managing critical business data and transactions

9. ETL (Extract, Transform, Load) - is a data process used to move and Prepare data from different Sources into a central System like (a data warehouse) for Analysis)

a, Extract - Process of gathering or retrieving raw data from various Source Systems

b, Transform - raw data is Cleaned, validated & Converted into a consistent, analytics-ready format. Include - deduplicating records, and aggregating metrics

c, Load - The final Step where the clean, transformed data is moved into a target repository such as google cloud, data warehouse a Snowflake data platform



10. a, CSV (comma-Separated values) - A Simple text format where data is stored in rows and Columns, Separated by Commas

-> Use it when working with Small to medium datasets

-> When data needs to be easily read or shared between Systems, Best for basic Spreadsheet

eg. Exporting Sales data from

b, Json - A Semi-Structured format that stores data in key-value pairs and Supports nested Structures - (Javascript Object Notation)

- when data is hierarchical (or Complex)

- useful when flexibility is needed

eg. Storing user profiles or data from web Services

c, Parquet - A Column based storage format designed for processing big data

eg. Processing large data sets in Cloud platforms like AWS or Google

d, Avro - A row-based data format that stores data along with its schema. for data streaming and data exchange between Systems.

eg Real time messaging systems like whatsapp

Section B: Artificial Intelligence & Machine learning

11. Artificial intelligent - Is the branch of Computer Science that focuses on Creating systems a machine that Can perform tasks that normally require human intelligen

a, Narrow AI (Weak AI) - designed to perform a Specific task or a limited set of tasks
 * It cannot think beyond what it is trained for eg Siri or Google assistant

b, General AI (Strong AI) - Able to understand, learn & apply intelligence across a wide range of tasks, Just like a human - eg a robot that Can think, learn & Perform

12. Machine learning - A branch of AI where Computers are trained to learn Patterns from data & make predictions or decisions without being Explicitly ~~trained~~ programmed for every task

- Traditional Programming - humans write fixed rules & instructions, the computer follows those rules to produce output

3 types of machine learning

1. Supervised learning	2. Unsupervised learning	3. Reinforcement Learning
- Model learns from input data that has correct output labels it learns a mapping from inputs to outputs & make Predictions example: Spam dection, image recog, house Price predictions	- The model works with data that has no labels: discovers hidden Patterns, structures or Groups in the data on it's own eg Customer Segmentation	- The model learns by interacting with environment & receiving Feedback in form of rewards or Penalties - Eg training a robot to walk, chess, Autonomous Navigation

13. Deep learning - Uses artificial neural networks to learn from vast amounts of unstructured data - allows computers to learn in a way that is human brain inspired

AI - Artificial Intelligence	Machine Learning
- Refers to any system that can Perform ~~tasks~~ that normally require human intelligence like thinking, learning & decision making	- It focuses on systems that learn from data instead of being explicitly Programmed

*

Neural Networks - Computing systems inspired by the structure & function of human brain
- They are key functions of Machine learning & deep learning and are used to recognise patterns in data
- A neural network - is made up of connected units called neurons (nodes) that works together to process information.
- Consist of 3 parts:
① Input layer
* Recieves data
② Hidden layers
Process & learn Patters in data
③ Output layer - Produces the final result (Prediction or Classification)

14. Natural Learning Processing (NLP) - A branch of AIs that focuses on enabling Computers to understand, interpret & generate human language eg Venda, Tsonga
* NLP allows computer to understand written texts (emails, Messages, documents)
* Translate languages (eg English, Zulu)
* Generate human-like texts (chatbots), AI Assistants
eg when you ask chatgpt, Siri Questions NLP is used to understand your language
Google translate

15. Large language Model - A type of AI model trained on Massive amounts of text data to understand, Generate & respond in human language
two well know LLMs
* chatgpt (OpenAi)
- Used for Conversational AI & text generation tasks, It can understand Questions & Provide answers
* Gemini (Google)
- Used for advanced reasoning, Search assistant and Content Generation

16. Generative AI - type of AI that can create new content such as text, images, Audio video, or code based on patterns it has learned from data.
- it learns Patterns from large datasets then Generate new Content
① ChatGPT - Generate text. Create human like text responses
② Dall-E - Generates Images - Creates images from text description
③ Github Copilot - Generates Codes - helps write Computer Codes

17. Training dataset - A collection of data used to teach a machine learning or AI model how to make predictions or decisions.
- Quality is important because AI model can only learn from data its given
- Poor data, poor results; incorrect data leads to inaccurate results

18. Computer vision - A field in AI that enables computers to see, interpret and understand images & videos in a way similar to humans
- Machines dont see like humans they interpret images as numbers (pixels)
examples
① facial recognition - Smart phones & Security Systems
- As used in phone face locks systems & airport Security
② Automomous vehicles (self driving cars)
- Cv used in cars like Tesla to detect pedestrians, road signs, lane
- Helps the car make safe driving decisions in real time

19. Section C: APIs, Tools & Developer Concepts

19. API (Application Programming interface) - Is a set of rules that allows different Software systems to communicate with each other
eg Weather App - to get real-time weather data from data services
- logging in on FB through another app, API are used to connect services
- API allows different APPS to communicate, share data and work together efficiently
- Save development time - developers can use existing services instead of starting from zero

20. API key - is a unique code (like password or digital ID) used to identify & Authenticate a user, application or system when it tries to access an API
- API key can be a string of letters & numbers given by a service to control who can use their API
- it confirms the the request is coming from the valid user or Application

21. Rest API - A type of web API that allows different systems to communicate over the internet using standard HTTP methods like GET, POST, Put & delete
· REST - Representational State transfer

22. Json - is a light weight text-based format used to store & Exchange data, organizes data in key-value pairs
why commonly used with APIs & data exchange
- It is easy to read & write - Simple structure
- uses less data - lightweigh
- language indipendent - works with almost any programming language

23. Prompt Engineering - is the skill of designing clear & effective instructions for AI tools like ChatGPT or Anthropic's Claude to get accurate, useful, and relevant responses

Examples of Poorly writen Prompt
1) Developing an app to advertise my company that Sells Sweet potatoes
- AI doesn't know target market, if not given such information the website can be poorly designed
Improved prompt: Give AI all the specifics you want to see

24. Map function - A programming function used to apply the same operation to every item in a list & return a new list with transformed result
- it takes a collection of data, applies a function to each item and returns a new collection with the updated values

25. Cloud Computing - access of powerful computer resources online without physically owning physical hardware
a) IaaS (infrastructure as a service) - Amazon
b) PaaS (Platform as a service) - Google cloud
c) SaaS (Software as a service) - Microsoft 365

26. Version Control System (VCS) - tools that helps track & manages changes to files
Git - widely used distributed version Control System that tracks changes in code
Commit - Changes or Saves the changes made to a file
Branch - is a separate line of development within a project

27. Jupter notebook - interactive web-based tools used to create and share documents that contain live code, text, visualizations & output results
- Commonly used with Python
Popular among data Analyst and Scientist
* It is iteractive, Great for data exploration, combines codes & explanation

28. Python - is a programming language most widely used in data analytics and artificial intelligence (AI). It is easy to learn & read, has powerful libraries, large community support
4 Commonly used libraries
1. Pandas - Data manipulation & Analysis
- Structured data
- Helps clean, filter & transform data
2. Numpy - Used for numerical computing
- handles arrays & Matrices

3. Matplotlib - used for data visualization
- Creates Graphs like plots, bar charts & Scatter plots

4. Scikit - learn - Provides tools for building AI models
- Supports classification, regression, clustering and model evaluation

Section D

29. Data Privacy - Protection of Personal Information & How it is collected, stored, used & Shared by organizations
It is import because it protects individuals from harm, build trust
Prevents data misuse
GDPR - (General Data Protection Regulation) - Is a data Protection law Created by the European Union, that came in effect May 2018
- Applies to European Contries such as France
Germany
Italy
Spain
Netherlands

30 Data Ethics - Moral Principles & Guidelines that Govern how data is collected, stored, shared & used. Especially when it involves AI systems
Examples of data misuse
① Biased AI decision making systems → Hiring system favouring one Group
② Suirvellance without transparency or Accountability - Facial recognition without consent

③1 Data Governance - Polices, processes, roles, & standards that ensures an organization's data is accurate, secure, consistent & used responsibly
Key concepts of good data Governence
A. Data Quality management
b) Data Polices & Standards
c) Data Sequrity & Privacy

32. Bias in AI - Systematic & unfair errors in artificial intelligence System that produce results favouring one Group
- AI model biased - Is when AI is trained on data that is not Representative of the real world
eg when AI is trained on that with inequality, hiring more men on engineering job for CV shortlisting

~~33. Business Intelligence~~
33.

	Roles	Skills	Tools
Data Engineering	building & maintaining data	data warehousing ~~Advances Stats~~	Python, ~~R~~, SQL, ~~Jupyter~~
Data Analyst	Intepreting historic data	Data Cleaning	SQL Excel, tableau
Data Scientist	finding hidden Patters	Advanced Stats	Python, R, Jupyter
AI Engineer	deploying Scaling & integrating AI	Software Eng	Python, face, hugging Pytorch

34 Business Intelligence - Is the process of taking raw, messy business data and turning it into clear, actionable insights
a. Microsoft Power BI - Generate automated dashboard for sales daily (can see stores underperforming), so business can make adjustment
b. Tableau - Discover or decision makers can interact with a Tableau Dashboard to filter data by specific hours of the day

35. Data Ethics & Governance
- it is interesting to know there are laws and policies that governs the whole data science & Analysis field

36. The python libraries mostly use for data, I watched Youtube to help me understand those concepts

37. As a research Scientist, Data & AI can support my career. e.g it can analyse large biological datasets much faster than doing it the tradional way. It can help identify disease targets patterns. Predict outcomes & improve drug or Experiments reducing human labour, error and time

Reference

1. Youtube
2. AWS
3. Chatgpt
4. Gemini